

This fictional city brief was created for teaching RAG retrieval and evaluation.

Rivergate is a fictional river city with low neighborhoods along the Alder River. The city council identifies spring floods, storm surge from the lower basin, and overloaded drainage pipes as the main climate risks. The highest exposure is in Alder Quay, Millbank, and the Canal View senior housing area.

The first priority is a blue-green corridor east of Millbank. The plan restores 18 hectares of wetlands and reconnects two old side channels to slow flood water before it reaches homes. Engineers expect the corridor to reduce peak water levels during medium floods and create temporary storage during heavy rain.

Rivergate will also install removable flood barriers along Alder Quay. The barriers are not permanent walls. They are stored near the quay and deployed when the river forecast crosses the orange alert threshold. The city chose removable barriers because businesses wanted daily access to the waterfront.

The monitoring program adds 42 smart river gauges and rain sensors. The gauges report water level, flow speed, and blocked drain warnings every 10 minutes. Emergency managers use this data to decide when to close the quay road and when to open evacuation route R-3 toward North Hill School.

The social vulnerability section gives special attention to Canal View senior housing. During high flood alerts, volunteers check ground-floor apartments, arrange accessible transport, and move medicine storage to the second floor community room. The plan says that flood protection is not successful if vulnerable residents cannot act on warnings.

Phase one has a budget of 7.8 million city credits over three years. The target is to keep the community center, the clinic, and the main grocery store dry during a 1-in-50-year river flood. The plan does not claim to remove all risk. It aims to reduce damage, improve warning time, and make evacuation decisions easier.